NFL Position Level Draft Prediction Modeling

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Introduction/Abstract

Each year, approximately 250–260 of the 329 athletes invited to the National Football League (NFL) Combine are drafted into the league. To get invited, you must have an exceptional track record in your respective collegiate career. The purpose of the NFL Combine is to collect additional data on these top prospects through one-on-one team interviews, cognitive evaluations, and physical tests, providing teams with the information needed to assess organizational fit. The goal of our project is to determine how well a player's performance metrics and college production can predict NFL draft if they are drafted or not. Our analysis is limited to these three positions: quarterback, wide receiver, and running back. This project evaluates the performance of tuned multilayer perceptrons and Bayesian logistic regression models in predicting whether a player is drafted, and identifies the most important variables driving these predictions.

Motivation and Data

Our data was procured from two primary sources and joined based on player name and position. These are linked here: **"NFL Combine Dataset"** and **"College Football API"**. The first is an open-source NFL Combine performance dataset containing 4,741 entries from 2010 through 2023. The second dataset is sourced from the 'College Football Data' (CFB) API, which provides comprehensive yearly statistics for all college football players. For reference, the NFL Combine dataset includes both physical characteristics and performance metrics such as height, weight, 40-yard dash time, vertical jump, broad jump, three-cone drill time, and bench press repetitions. The CFB dataset, in contrast, contains college performance statistics, including metrics like passing yards or receptions for a given player in a specific season.

Pictured: an infographic showing the steep decline in the number of athletes who progress through each level beyond high school football, underscoring how difficult it is to reach the NFL.

To merge these datasets, our data cleaning process first pivots the CFB data by player name and retains only the most recent season for each player. This ensures that our dataset includes only the most relevant metrics from a player's final year of eligibility. Our matching process applies fuzzy string matching with an 88.0% similarity threshold on both the cleaned player name and standardized position to identify corresponding records. Even with our normalization and matching procedures, our approach achieves a 52.9% match rate on our Combine data when performing an inner join between the Combine dataset and the CFB dataset.

Methodology

Our central question is whether we can predict, for each of the three offensive positions, if a player will be drafted based on combine results and final season college football statistics. We implemented two models. The first was a multilayer perceptron. For this model, we min-max scaled the data, applied an 80/20 train-test split, and used a two-layer hidden layer architecture with ReLU activations. To address class imbalance, we incorporated class weights, and we trained the model using binary cross-entropy loss. Our second model was a Bayesian logistic regression model. We modeled draft status as a Bernoulli outcome with a logistic link function and placed normal priors on the regression coefficients. We then used Metropolis-Hastings sampling to estimate the posterior distribution of the model parameters, running the sampler for 25,000 iterations with a 20% burn-in period and thinning.

Results

After developing our two models for the three offensive positions and analyzing the results, both the multilayer perceptron and Bayesian logistic regression models showed their own strengths and weaknesses. The MLP achieved slightly better test accuracy for two of the three positions, with 60.5% for quarterbacks, 67.2% for wide receivers, and 61.8% for running backs, compared to 55.8%, 65.2%, and 63.2% for the Bayesian logistic regression model. However, F1 scores were relatively similar across both models, with no clear overall advantage. Overall, the wide receiver position had the strongest predictive performance, while the quarterback was the weakest for both approaches. Although the MLP performed better in terms of accuracy in most cases, the Bayesian model achieved higher F1 scores in two positions, suggesting a better balance between precision and recall. Both models indicate that the quarterback position is the hardest to predict accurately. This is not surprising given that quarterbacks have quite a large number of factors that contribute to their success or failure. For both models, the wide receiver predictions were the strongest, though there were still notable prediction outliers, such as the Bayesian missclassification of Cooper Kupp, the 2021 Triple Crown winner.

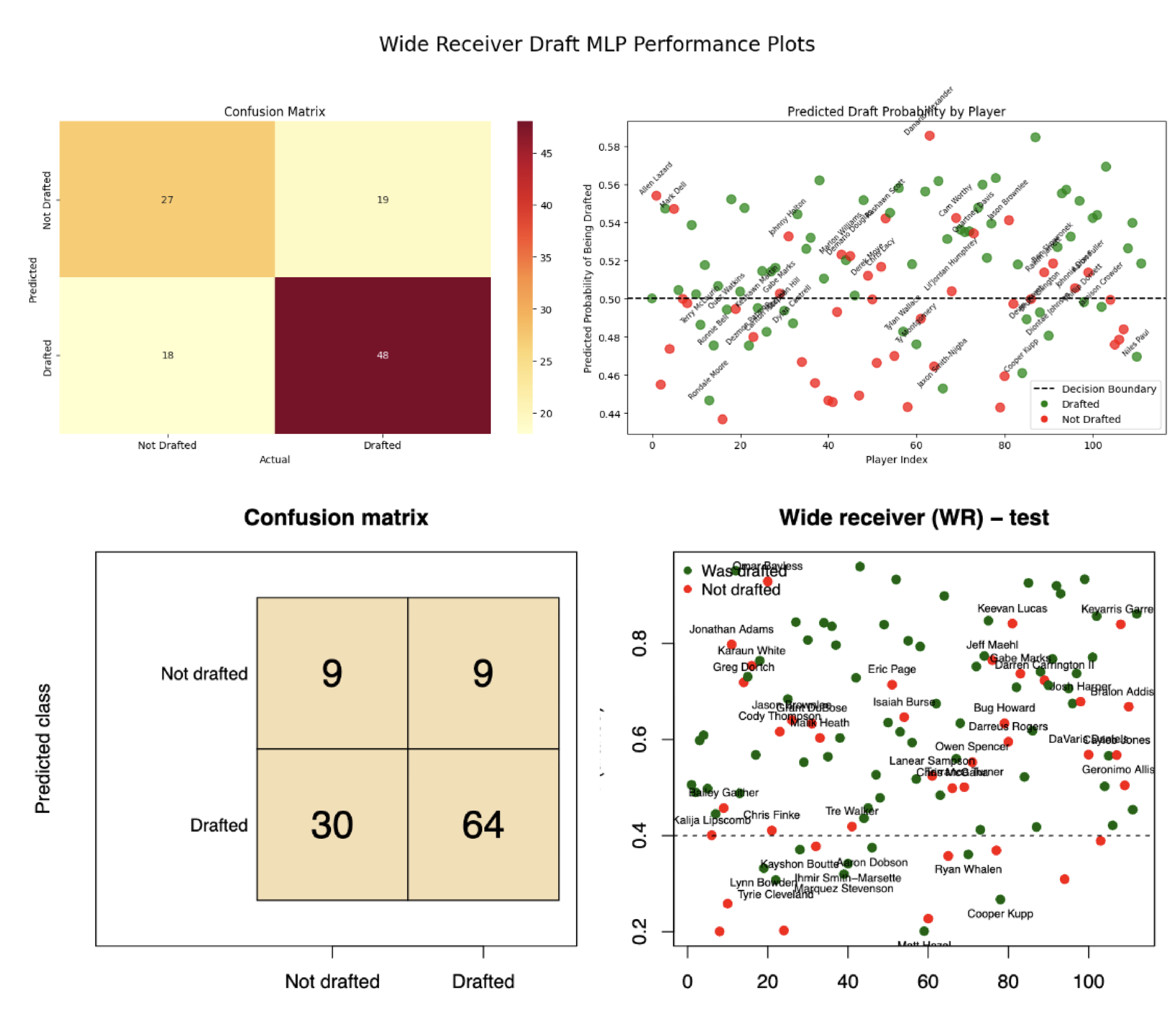

Fig. 1: Confusion matrices and prediction plots (top: MLP, bottom: Bayesian Logistic Regression)

Results Continued

Position	MLP Accuracy	MLP F1	Bayesian Accuracy	Bayesian F1
QB	60.5%	0.721	55.8%	0.678
RB	61.8%	0.705	63.2%	0.713
WR	67.0%	0.722	65.2%	0.766

Fig. 2: Test set accuracy and F1 scores for MLP and Bayesian Logistic Regression models

Discussion/Future Work/Web App

Overall, both models performed reasonably well and produced similar predictive results, with wide receivers being the most predictable position for draft outcomes and quarterbacks being the least predictable. Considering possible improvements and extensions for this project, a significant limiting factor was the small position dataset sizes observed after matching the two datasets. To address this, it may make sense to go through a more intensive manual matching process in order to train and test our models with more comprehensive datasets. There were a significant number of null values in our combined dataset, so adding either item or column-based collaborative filtering approaches would likely provide us with a more comprehensive dataset. For future work, we would likely want to procure more historical data in order to improve the training ability of both models. Our outliers also provided some insight that our predictions could benefit from additional metrics, including injury history, college strength of schedule, or college team division.

Project Website Link

References

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